

Amin Karimi Monsefi

PH.D. STUDENT OF COMPUTER SCIENCE.

Columbus, Ohio, USA

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“Be the change that you want to see in the world.”

Summary

Amin Karimi Monsefi is a Ph.D. candidate in Computer Science at The Ohio State University, advised by Professor Rajiv Ramnath. His research advances **efficient and controllable generative models** and the **representations** that power them, organized around three threads: (i) **few-step diffusion and flow matching** for vision and language (FS-DFM, TaxaDiffusion, DiReCT, KnobGen); (ii) **self-supervised and vision-language representation learning** for fine-grained understanding (FOLK, DetailCLIP); and (iii) **applied generative and representation learning** in high-impact scientific domains, including biodiversity (TaxaDiffusion, TaxaAdapter, SeamCam), 3D medical imaging (Masked LoGoNet), and multi-modal spatiotemporal prediction (CrashFormer). In Summer 2025 he interned at **Apple** on the MIND team, developing few-step discrete diffusion and flow-matching methods for large-scale text generation. His work has been published at top venues including **ICLR (2025, 2026), ICCV 2025, and SIGKDD (2023, 2024)**, accumulating over **375 citations**. He was recently honored with the **Graduate Research Award** from the OSU CSE Department (2026). He also serves as a reviewer for CVPR, ECCV, ICLR, ICML, SIGKDD, and WACV, recognized as an **Outstanding Reviewer (top 10%)** at KDD 2025 and an **Excellent Reviewer** at KDD 2026.

Education

PhD. in Computer Science

Columbus, Ohio, USA

THE OHIO STATE UNIVERSITY

Jan. 2022 - Exp. Present

- **Supervisor:** Prof. Rajiv Ramnath
- **Research focus:** Efficient and controllable diffusion-based generative modeling across continuous and discrete spaces, together with the representations that support it.
- **(1) Diffusion & Flow Matching in Continuous Space.** Developing controllable and physics-aware generative models for images and video, with few-step samplers that approach or surpass thousand-step teachers. Contributions include physics-refined video generation via disentangled contrastive trajectories (*DiReCT*), progressively trained fine-grained image generation across the tree of life (*TaxaDiffusion*, ICCV 2025; *TaxaAdapter*), and sketch-conditioned artwork synthesis with adjustable sophistication (*KnobGen*, CVPR-CVEU 2025).
- **(2) Discrete Diffusion & Flow Matching for Language.** Designing fast and accurate diffusion language models that close the gap with autoregressive systems while remaining parallel and bidirectional. Contributions include step-aware discrete flow matching for long text generation (*FS-DFM*, ICLR 2026), energy-navigated trajectory distillation that enables an 8-step student to surpass its 1024-step teacher (*TS-DFM*), and denoising-aware per-step credit assignment for RL on diffusion LMs (*DACA-GRPO*).
- **(3) Self-Supervised & Vision-Language Representation Learning.** Designing pretraining objectives that capture fine-grained structure, including frequency-guided masked image modeling (*FOLK*, ICLR 2025) and detail-oriented contrastive vision-language learning (*DetailCLIP*, ICLR-SSI-FM 2025). These representations underpin downstream generation, recognition, and segmentation tasks.
- **(4) Applied Generative & Representation Learning for Scientific Domains.** Translating the methods above to high-impact applications: camouflage quantification (*SeamCam*), self-supervised 3D medical image segmentation (*Masked LoGoNet*, KDD 2024), and multimodal spatiotemporal prediction for smart mobility (*CrashFormer*, SIGSPATIAL 2023).

M.Sc. in Computer Engineering with focus on Software

Tehran, Iran

SHAHID BEHESHTI UNIVERSITY

Sep. 2015 - Exp. Sep. 2017

- **Thesis:** An Approach for Automatic Software Test Data Generation Using Machine Learning and Program Static Structure
- **Supervisor:** Prof. H.Haghighi
- I achieved **Ranked 1st**, National M.Sc Entrance Exam of Computer Engineering, Among More Than 20,000 Students

B.Sc. in Computer Engineering with focus on Hardware

Tehran, Iran

SHAHID BEHESHTI UNIVERSITY

Sep. 2011 - Exp. Sep. 2015

- **Thesis:** Applying Reinforcement Learning on Multi-Agent Environments
- **Supervisor:** Prof. M. Abdoos

Papers

Trajectory as the Teacher: Few-Step Discrete Flow Matching via Energy-Navigated Distillation

NeurIPS

AMIN KARIMI MONSEFI, DOMINIC CULVER, NIKHIL BHENDAWADE, MANUEL CIOSICI, YIZHE ZHANG, IRINA BELOUSOVA

2026

DACA-GRPO: Denoising-Aware Credit Assignment for Reinforcement Learning in Diffusion Language Models

NeurIPS

AMIN KARIMI MONSEFI, DOMINIC CULVER, NIKHIL BHENDAWADE, LOKESH BOOMINATHAN, MANUEL CIOSICI, YIZHE ZHANG, IRINA BELOUSOVA

2026

TaxaAdapter: Vision Taxonomy Models are Key to Fine-grained Image Generation over the Tree of Life

NeurIPS

MRIDUL KHURANA, AMIN KARIMI MONSEFI, JUSTIN LEE, DAVID CARLYN, JULIA CHAE, JIANYANG GU, RAJIV RAMNATH, SARA BEERY, WEI-LUN CHAO, ANUJ KARPATNE, CHENG ZHANG

2026

Controlla: Learning Controllability via Graph-Constrained Latent Geometry

Under Submission

JAMUNA S MURTHY, AMIN KARIMI MONSEFI, RAJIV RAMNATH

2026

KathaTrace: Diagnosing Semantic Trajectory Collapse in Generated Visual Narratives

Submitted for WACV

JAMUNA S MURTHY, AMIN KARIMI MONSEFI, RAJIV RAMNATH

2026

SeamCam: Quantifying Seamless Camouflage via Multi-Cue Visual Detectability

Under Submission

AMIN KARIMI MONSEFI, ABOLFAZL MEYARIAN, MRIDUL KHURANA, BRANT WANG, POUYAN BORESHNAVARD, ANUJ KARPATNE, CHENG ZHANG, WEI-LUN CHAO, RAJIV RAMNATH

2026

DIRECT: Disentangled Regularization of Contrastive Trajectories for Physics-Refined Video Generation

Submitted for WACV

AMIN KARIMI MONSEFI, ABOLFAZL MEYARIAN, RAJIV RAMNATH, SER-NAM LIM

2026

FS-DFM: Fast and Accurate Long Text Generation with Few-Step Diffusion Language Models

ICLR

AMIN KARIMI MONSEFI, NIKHIL BHENDAWADE, MANUEL RAFAEL CIOSICI, DOMINIC CULVER, YIZHE ZHANG, IRINA BELOUSOVA

2026

- The Fourteenth International Conference on Learning Representations
- Rio de Janeiro, Brazil

ISOSNet: a unified framework for cone photoreceptor detection and inner segment and outer segment length measurement from AO-OCT B-scans

Biomedical Optics Express

MENGXI ZHOU, YUE ZHANG, ELI KIRKENDALL, AMIN KARIMI MONSEFI, MATTHEW WOLFE, KIRAN A CHITKARA, STACEY S CHOI, NATHAN DOBLE, SRINIVASAN PARTHASARATHY, RAJIV RAMNATH

2025

- Biomedical Optics Express is Optica Publishing Group
- Journal Biomedical Optics Express
- Impact Factor: 2.9

TaxaDiffusion: Progressively Trained Diffusion Model for Fine-Grained Species Generation

ICCV

AMIN KARIMI MONSEFI, MRIDUL KHURANA, RAJIV RAMNATH, ANUJ KARPATNE, WEI-LUN (HARRY) CHAO, CHENG ZHANG

2025

- International Conference on Computer Vision
- Honolulu, Hawai'i, USA

KnobGen: Controlling the Sophistication of Artwork in Sketch-Based Diffusion Models

CVEU Workshop - CVPR

AMIN KARIMI MONSEFI, POUYAN BORESHNAVARD, MENGXI ZHOU, WEI-LUN (HARRY) CHAO, ALPER YILMAZ, RAJIV RAMNATH

2025

- CVPR - AI for Creative Visual Content Generation Editing and Understanding (CVEU) Workshop
- Nashville, Tennessee, USA

Frequency-Guided Masking for Enhanced Vision Self-Supervised Learning

ICLR

AMIN KARIMI MONSEFI, MENGXI ZHOU, NASTARAN KARIMI MONSEFI, SERNAM LIM, WEI-LUN (HARRY) CHAO, RAJIV RAMNATH

2025

- The Thirteenth International Conference on Learning Representations
- Singapore

DetailCLIP: Detail-Oriented CLIP for Fine-Grained Tasks

SSI-FM Workshop - ICLR

AMIN KARIMI MONSEFI, KISHORE PRAKASH SAILAJA, ALI ALILOOEE, SER-NAM LIM, RAJIV RAMNATH

2025

- ICLR - Scaling Self-Improving Foundation Models (SSI - FM) Workshop
- Singapore

Masked LoGoNet: Fast and Accurate 3D Image Analysis for Medical Domain

KDD

AMIN KARIMI MONSEFI, PAYAM KARISANI, MENGXI ZHOU, STACEY CHOI, NATHAN DOBLE, HENG JI, SRINIVASAN

2024

PARTHASARATHY, RAJIV RAMNATH

- 30TH ACM SIGKDD - Conference On Knowledge Discovery And Data Mining
- Barcelona, Spain

Reducing manual labeling requirements and improved retinal ganglion cell identification in 3D AO-OCT volumes using semi-supervised learning

Biomedical Optics Express

MENGXI ZHOU, YUE ZHANG, AMIN KARIMI MONSEFI, STACEY S CHOI, NATHAN DOBLE, SRINIVASAN PARTHASARATHY, RAJIV

2024

RAMNATH

- Biomedical Optics Express is Optica Publishing Group
- Journal of Biomedical Optics Express
- Impact Factor: 2.9

CrashFormer: A Multimodal Architecture to Predict the Risk of Crash

31st ACM SIGSPATIAL

AMIN KARIMI MONSEFI, POUYA SHIRI, AHMAD MOHAMMADSHIRAZI, NASTARAN KARIMI MONSEFI, RON DAVIES, SOBHAN

2023

MOOSAVI, RAJIV RAMNATH

- UrbanAI '23: Proceedings of the 1st ACM SIGSPATIAL
- Hamburg, Germany

Novel Physics-Based Machine-Learning Models for Indoor Air Quality Approximations

KDD

AHMAD MOHAMMADSHIRAZI, AIDA NADAFIAN, AMIN KARIMI MONSEFI, MOHAMMAD H RAFIEI, RAJIV RAMNATH

2023

- 29TH ACM SIGKDD - Conference On Knowledge Discovery And Data Mining
- Long Beach, California, USA

Smart and collaborative industrial IoT: A federated learning and data space approach

Elsevier - Journal

BAHAR FARAHANI AND AMIN KARIMI MONSEFI

2023

- Journal: Digital Communications and Networks
- Impact Factor: 7.5

Will there be a construction? Predicting road constructions based on heterogeneous spatiotemporal data

30th ACM SIGSPATIAL

AMIN KARIMI MONSEFI, SOBHAN MOOSAVI AND RAJIV RAMNATH

2022

- Conference: International Conference on Advances in Geographic Information Systems
- Seattle, Washington, USA

Professional Experiences

Apple

Seattle, WA, USA

ML RESEARCH INTERN

May 2025 - Present

- Apple MIND Team. Research focus: fast and principled generative modeling with discrete diffusion and flow matching.
- Developed **FS-DFM**, a step-aware discrete flow-matching framework that matches the quality of 1024-step diffusion baselines in just 8 steps (128x speedup), outperforming LLaDA-8B and Dream-7B despite being 40x smaller. [ICLR 2026]
- Designed a reinforcement learning framework (**DACA-GRPO**) for diffusion language models, introducing per-step credit assignment and stratified likelihood estimation that improve reasoning on MATH-500, GSM8K, and Sudoku with zero extra inference cost. [Submitted for NeurIPS 2026]
- Proposed **trajectory shaping** for discrete flow distillation: an energy-guided navigation scheme and a contrastive curriculum that enable the distilled student to surpass its 1024-step teacher, with all improvements baked into weights at training time. [Submitted for NeurIPS 2026]

Higharc

North Carolina, USA

MACHINE LEARNING INTERN

May 2024 - Aug 2024

- Conducting research on semantic and panoptic segmentation tasks.
- Utilizing unlabeled data to pre-train a DETR-based model and addressing challenges of limited labeled data with Self-Supervised learning.
- Implementing domain adaptation approaches to generalize models across multiple datasets with distinct distributions. Also, developing strategies to transfer a trained model from one domain to another effectively.

JIBB

San Francisco, USA

SENIOR MACHINE LEARNING ENGINEER

Dec 2020 - Dec 2021

- Detected and removed unwanted objects from images and videos using advanced object detection techniques.
- Implemented methods to detect colors, remove shadows, and eliminate reflections from images and videos.
- Developed and optimized various U-Net-based architectures to enhance image and video processing tasks.
- Utilized machine learning algorithms and computer vision techniques for real-time image and video enhancement.

CTO

Dec 2018 - Dec 2021

- Served as the technical consultant, providing strategic guidance and oversight for technical projects.
- Designed high-level architecture for pharmacy software solutions, ensuring scalability, reliability, and efficiency.
- Collaborated with cross-functional teams, including software developers and business experts, to align technical strategies with business goals.
- Implemented best practices in software engineering, including agile methodologies and continuous integration/continuous deployment (CI/CD) pipelines.
- Led efforts to enhance software performance and security, ensuring compliance with industry standards and regulations.

TAPSI (Online Taxi)

Tehran, Iran

SENIOR DATA SCIENTIST & BACK-END DEVELOPER

Mar. 2018 - Dec 2020

- Designed and implemented pricing micro-services using Python, communicating through event messaging with RabbitMQ.
- Developed a fake GPS detection algorithm to prevent fraud and have a safe and secure system.
- Created intelligent backend features based on passengers' data, including origin suggestion, destination suggestion, favorite location suggestion, and ride suggestion using unsupervised.
- Implemented a micro-service for estimated arrival time (ETA) using drivers' GPS information. Introduced a new algorithm for ETA based on GPS data and published a paper on this work. I also published a paper in this manner.
- Developed a smart map for predicting locations with higher ride demand rates in cities.
- Utilized Python and Node.js for back-end development, Spark, Kafka, and R for analytical purposes, and PostgreSQL, MongoDB, and Redis.
- Python, NodeJS, Spark, Kafka, RabbitMQ, PostgreSQL and MongoDB

Honors & Awards

RESEARCH RECOGNITION

2025-26 **Graduate Research Award**, Selected by the Department of Computer Science and Engineering, The Ohio State University, for distinguished research contributions in generative modeling. Columbus, Ohio

REVIEWER SERVICE & RECOGNITION

KDD 2026 **Excellent Reviewer**, Second highest grade among reviewers. Jeju, Korea
 KDD 2025 **Outstanding Reviewer**, Second round - 10% of top reviewers received. Toronto, Canada
 KDD 2025 **Excellent Reviewer**, First round - 20% of top reviewers received. Toronto, Canada

ACM-ICPC COACH AND TEAM LEAD

Apr 2018 **Coach**, 42nd Annual World Finals Beijing, China
 Dec 2017 **Coach**, 7th and 17th teams, Regional Contest Tehran, Iran
 Nov 2017 **Coach**, 10th and 16th teams, Internet Programming Contest Tehran, Iran
 Oct 2017 **Coach**, The 1st team, Regional Contest Kabul, Afghanistan
 Oct 2017 **Coach**, The 1st team, Online Programming Contest Kabul, Afghanistan
 Oct 2015 **Coach**, The 11th team, Iran-Internet-PC-IU Online Programming Contest Tehran, Iran
 Oct 2015 **Coach**, The 23rd team, Iran-Internet-PC-SBU Online Programming Contest Tehran, Iran
 Nov 2014 **Coach**, The 12th, 53rd, and 131st teams, Internet Programming Contest Tehran, Iran

ACM-ICPC CONTESTANT

Dec 2014 **10th Place**, Asia Regional Contest Tehran, Iran
 Oct 2014 **2nd Place**, Qualification Contest, Shahid Beheshti University Tehran, Iran
 Dec 2013 **3rd Place**, Asia Regional Contest Tehran, Iran
 Oct 2013 **1st Place**, Qualification Contest, Shahid Beheshti University Tehran, Iran
 May 2013 **5th Place**, The 5th ACM National Collegiate Programming Contest, University of Kashan Kashan, Iran
 Apr 2013 **8th Place**, The 1st ACM National Collegiate Programming Contest, University of Shahrood Shahrood, Iran
 Dec 2012 **25th Place**, Asia Regional Contest Tehran, Iran
 Nov 2012 **16th Place**, The 3rd ACM National Collegiate Programming Contest, Amir Kabir University of Technology Tehran, Iran
 Oct 2012 **2nd Place**, Qualification Contest, Shahid Beheshti University Tehran, Iran

OTHER

June 2015 **Ranked 1st**, National M.Sc Entrance Exam of Computer Engineering, Among More Than 10.000 Students Iran
 June 2011 **Ranked Top 1%**, National B.Sc Entrance Exam of Mathematics, Among More Than 256,000 Students Iran
 Mar 2009 **Bronze Medal**, Mathematics Olympiad of Waterloo University Tehran - Waterloo

Teaching Experiences

TEACHING ASSISTANT

2 times	AI for Non-Majors , Instructors: Ali Alilooee (Spring 25) and Mike Green (Fall 26)	Ohio State Uni
Spr 2025	Computer Vision for Human-Computer Interaction , Teacher: Dr. Wei-Lun (Harry) Chao	Ohio State Uni
5 times	Modeling and Problem Solving with Spreadsheets and Databases , Fall 22/23/25 and Spring 23/24	Ohio State Uni
Spr 2012	Introduction to Programming , Teacher: Dr. A. M. Aznaveh	Shahid Beheshti Uni
Fall 2012	Introduction to Programming , Teacher: Dr. A. M. Aznaveh	Shahid Beheshti Uni
Spr 2013	Introduction to Programming , Teacher: Dr. M. H. Moayeri	Shahid Beheshti Uni
Spr 2013	Discrete Mathematics , Teacher: Dr. F. Safaie	Shahid Beheshti Uni
Fall 2013	Discrete Mathematics , Teacher: Dr. H. Haghighi	Shahid Beheshti Uni
Spr 2014	Discrete Mathematics , Teacher: Dr. F. Safaie	Shahid Beheshti Uni
Spr 2015	Introduction to Algorithm , Teacher: Dr. E. Bavafaye	Shahid Beheshti Uni
Fall 2015	Data Structure , Teacher: Dr. M. E. Moghadam	Shahid Beheshti Uni
Fall 2015	Artificial Intelligence , Teacher: Dr. M. Shamsfard	Shahid Beheshti Uni
Fall 2015	Introduction to Algorithm , Teacher: Dr. R. Ghavamizade	Shahid Beheshti Uni
Fall 2016	Data Structure , Teacher: Dr. M. E. Moghadam	Shahid Beheshti Uni
Fall 2016	Artificial Intelligence , Teacher: Dr. M. Shamsfard	Shahid Beheshti Uni
Spr 2017	Discrete Mathematics , Teacher: Dr. H. Haghighi	Shahid Beheshti Uni
Spr 2018	Discrete Mathematics , Teacher: Dr. H. Haghighi	Shahid Beheshti Uni